

Digital Image Processing

Homework 3

{Chromatic Adaptation + Image Enhancement + Color Temperature Adjustment}

I. Chromatic Adaptation

(1) Color Space Conversion:

Standard Red Green Blue (sRGB) color space is the most widely used format for screen images. It is designed to be perceptually uniform by aligning pixel values with human brightness perception. However, sRGB values are non-linear due to gamma encoding, which compresses the extensive range of real-world brightness into a smaller digital range. In contrast, the linear light color space represents light intensity in direct proportion to the actual light emitted, making it essential for accurate image processing operations such as blending, color correction, and filtering. Gamma correction plays a crucial role by facilitating precise processing in the linear domain while ensuring consistent display quality when converting images back to sRGB, resulting in a natural appearance on screens.

(2) Gray World Algorithm:

The gray world assumption is a straightforward yet effective technique for automatic white balance adjustment in image processing. It operates on the premise that a well-balanced image exhibits equal average intensities across the red, green, and blue channels, resulting in a neutral gray color. The algorithm consists of three key steps: first, computing the average intensity for each channel across all pixels; second, calculating scaling factors based on a gray value derived from these averages; and third, applying the scaling factors to adjust the pixel values, thereby balancing the channels and correcting color bias. This method offers several advantages, including the elimination of unnatural color casts caused by varying lighting conditions, ease of implementation due to its computational simplicity, and effectiveness for general-purpose image processing tasks.

Summary:

Color space conversion ensures mathematically accurate operations by working in linear space before converting back for display. The Gray World algorithm adjusts the RGB channels to neutralize color biases and achieve a natural-looking white balance.

II. Image Enhancement**(1) Gamma Correction:**

This technique is used to adjust image brightness and tonal distribution by applying a non-linear transformation to pixel values, ensuring they align with the way humans perceive brightness non-linearly. This process makes linear pixel intensities appear more natural and visually consistent with human perception. Pixel values, initially ranging from 0 to 255, are normalized to the range $[0, 1]$, after which the gamma correction formula is applied. A gamma value greater than 1 brightens the image, while a value less than 1 darkens it (using the reciprocal of the gamma value). The resulting values are then scaled back to the range $[0, 255]$ and clamped to 255 to prevent overflow.

(2) Contrast Adjustment:

Contrast modifies the difference between light and dark areas in an image to enhance visual clarity and emphasize details. The process involves calculating the average intensity for each color channel (red, green, and blue) and adjusting pixel values based on their deviation from this average. A contrast factor greater than 1 increases the difference, making bright areas brighter and dark areas darker, while a factor less than 1 reduces the contrast. Adjusted pixel values are clamped to the 0–255 range to avoid overflow or underflow. This technique is commonly used to improve underexposed or flat images, enhancing detail visibility and creating a balanced contrast for better visual appeal.

(3) Saturation Adjustment:

Saturation refers to the intensity or purity of a color, which determines its vibrancy or muteness. It is adjusted to control the overall color vibrancy in an image. To modify saturation, RGB values are first converted to HSV (Hue, Saturation, Value). In this model, Hue represents the type of color, Saturation indicates color intensity (ranging

from 0 for gray to 1 for pure color, calculated using the maximum and minimum RGB values), and Value denotes brightness. The saturation value is then scaled, with factors greater than 1 increasing intensity and factors less than 1 decreasing it. Finally, the modified saturation is used to recalculate RGB values by converting the updated HSV values back to RGB. This process can enhance dull images by amplifying color vibrancy or tone down oversaturated images for a more natural appearance.

Summary:

Gamma correction enhances the brightness of images by adjusting pixel intensity to match human perception, resulting in a more natural and visually cohesive appearance. Contrast adjustment increases the distinction between light and dark regions, which helps to reveal details and improve clarity, particularly in images that are underexposed or lack depth. Saturation adjustment manages the intensity of colors, enabling the enhancement of dull images with bright hues or the softening of overly vibrant ones, achieving a more aesthetically pleasing balance. When combined, these methods improve overall image quality by enhancing brightness, detail, and color accuracy.

For **output1_1.bmp** image:

Enhancement	Red Factor	Green Factor	Blue Factor
Gamma Correction	1.6	1.8	2.0
Contrast	1.4	1.5	1.6
Saturation	1.5		

For **output2_1.bmp** image:

Enhancement	Red Factor	Green Factor	Blue Factor
Gamma Correction	1.8	2.0	2.5
Contrast	1.0	1.3	1.5
Saturation	2.2		

For **output3_1.bmp** image:

Enhancement	Red Factor	Green Factor	Blue Factor
Gamma Correction	1.6	1.6	1.6
Contrast	1.3	1.1	1.2
Saturation	1.5		

For **output4_1.bmp** image:

Enhancement	Red Factor	Green Factor	Blue Factor
Gamma Correction	0.6	0.5	0.4
Contrast	0.9	0.7	0.8
Saturation	1.2		

III. Color Temperature Adjustment

(1) Warm Tone:

Warm tones infuse images with a reddish or orange tint, evoking the warmth of sunlight or fire. They enhance reds and reduce blues while keeping greens unchanged to maintain balance. This adjustment adds richness and vibrancy to the image, softening cooler, bluish tones for a cozier appearance. In real-world applications, warm tones are often used to enhance portraits with a soft, glowing look, create the golden hour effect in landscape photography, and evoke a sense of warmth in indoor scenes.

(2) Cool Tone:

Cool tones introduce a bluish or cyan tint to images, evoking the coolness of ice or water by enhancing blues and reducing reds while keeping greens unchanged to preserve natural balance. This adjustment diminishes warmth, creating a fresh, crisp atmosphere with intensified blue tones. Common applications include enhancing winter or underwater scenes, giving photos a clinical or futuristic look, and emphasizing calmness or detachment in artistic compositions.

Summary:

Both warm and cool tone adjustments are essential in photography and image editing to evoke specific moods and enhance visual storytelling.

Tone Type	Red Factor	Green Factor	Blue Factor
Warm	1.2	1.0	0.8
Cool	0.8	1.0	1.2